

Kunal Ajgaonkar

AI/ML Engineer

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EXPERIENCE

AI/ML Intern

Jun '26 — Present

Dassault Systèmes Solutions Lab Pvt. Ltd.

Pune, Maharashtra, India

- Develop and enrich RDF-based Product Lifecycle Management (PLM) Knowledge Graphs by integrating product structures, supplier data, engineering issues, and change-management workflows to enable end-to-end traceability and graph-based analytics.
- Design Python-based data generation, simulation, and graph-enrichment pipelines for engineering issues, change requests, change actions, part revisions, and lifecycle histories, creating realistic datasets for AI/ML and knowledge-graph applications.
- Develop a knowledge-graph AI agent for natural-language querying over enterprise PLM data, incorporating LLM-driven query generation, SPARQL validation, and graph reasoning; migrated and integrated the solution within an enterprise AI platform.

Data Science Intern

Jan '25 — Jun '25

Carico Systems Private Limited

India (Remote)

- Developed an employee data analytics system using Streamlit, ChromaDB, and LLM-based natural-language querying, with a scalable ChromaDB-SQLite synchronization architecture for large datasets and actionable insight generation.
- Led development of LogixSense, an AI-driven logistics platform integrating machine-learning capabilities with a modern web technology stack.

AI/ML Intern

Aug '24 — Sep '24

Suvidha Foundation

India (Remote)

- Applied machine-learning and data-analysis techniques to operational datasets, generating actionable insights and supporting data-driven decision-making across organizational initiatives.

EDUCATION

M.Tech in Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology
(GPA: 8.30)

Aug '25 — Present
Pune, Maharashtra, India

B.Tech in Computer Science (Artificial Intelligence and Data Science), SVKM's NMIMS STME
(GPA: 3.45/4.00)

Aug '21 — May '25
Navi Mumbai, India

PROJECTS

AstroGuard – Multi-Agent RAG System for Space Situational Awareness Risk Assessment [Link](#) Dec '25 — Mar '26

- Built a 4-agent debate-and-synthesis pipeline (decompose → parallel analyze → critique → refine → supervisor synthesis) over a hybrid vector (LanceDB) + knowledge-graph (NetworkX) retrieval layer, with citation-grounded structured outputs enforced via tool-calling schemas.
- Engineered for resilience on local open-weight LLMs (Qwen2.5, 4-bit quantized): six-strategy JSON repair/extraction layer, checkpointed batch runner with auto-topic discovery from graph centrality, and persistent memory feeding back into both query history and the knowledge graph.

Radio–Optical Galaxy Classification Using Deep Learning (ConvNeXt, EfficientNet, Ensemble Models) [Link](#)

Sep '25 — Mar '26

- Developed a 6-class radio+optical galaxy classifier using ConvNeXt, EfficientNet, ResNet, and ViT, achieving state-of-the-art performance with a weighted ensemble reaching 97.6% accuracy ($\pm 0.31\%$) and 0.999 AUC.
- Designed a rigorous 10-run evaluation pipeline with confidence intervals, per-class F1 scoring, and stability analysis, producing publication-quality benchmarks for astronomical morphology classification.

AdaptiveBeta – Regime-Aware Dynamic Beta Prediction & Portfolio Optimization [Link](#)

Jan '26 — Apr '26

- Built an end-to-end MLOps pipeline for dynamic beta forecasting on NIFTY50 using XGBoost (MAE 0.0391, 76.5% directional accuracy), LSTM, and Kalman Filter, with HMM-based market regime detection and walk-forward backtesting across 2022–2026.
- Implemented a regime-aware signal priority system (VIX, BetaVol thresholds) feeding a PyPortfolioOpt MVO engine, achieving best-in-class drawdown control (Max DD -17.36%) with full AWS deployment via CodeBuild/CodeDeploy, DVC, and MLflow experiment tracking.

ECO-REWIND – Spatio-temporal Counterfactual Modeling for Hurricane Impact on Coastal Wetlands [Link](#)

Jan '26 — Present

- Designed a spatio-temporal deep learning pipeline (UNet-Mamba + UNet-PatchTST) on 8-channel satellite patches (optical + SAR) to estimate counterfactual hurricane impact on coastal wetlands across the Florida Everglades and Barataria Bay.
- Engineered a robust preprocessing pipeline yielding 1,731 valid patches with optical-SAR validity masking, fp32-stabilized Mamba SSM scans to prevent float16 overflow, and per-band R² evaluation — enabling publication-ready counterfactual impact quantification for Hurricane Irma and Ida.

SSA-Intel – Named Entity Recognition for Space Situational Awareness Documents [Link](#) Jan '26 — Present

- Developed a 14-stage NER pipeline on ISRO/ESA technical documents using SciBERT and RoBERTa-base, achieving F1 = 0.9055 with BIO-schema label alignment, subword span continuation, and slug-aligned evaluation metrics.
- Resolved critical pipeline bugs across tokenizer alignment, BIO conversion, and span evaluation — producing a clean, retraining-free NER system for extracting orbital, debris, and mission entities from scientific literature.

SKILLS

Languages: Python, Java, C++, JavaScript, HTML, CSS
Machine Learning & Deep Learning: Scikit-learn, XGBoost, PyTorch, TensorFlow, CNNs, LSTMs, Transformers, Computer Vision, NLP
Generative AI & Knowledge Systems: LLMs, RAG, Agentic AI, Knowledge Graphs, RDF, SPARQL, ChromaDB, LanceDB, NetworkX
Data & MLOps: NumPy, Pandas, OpenCV, MLflow, DVC, AWS, CodeBuild, CodeDeploy
Tools & Platforms: Git, Streamlit, Power BI, Tableau, VS Code, Google Colab, LM Studio, Ollama

PUBLICATIONS

Hazardous waste management: A review on household hazardous waste and a strategy to gain control. Apr '24
STM Journals

Enhancing Cryptocurrency Fraud Detection with Hybrid Graph-Temporal Neural Networks Dec '24
IEEE

CERTIFICATIONS

MLOps with AWS Bootcamp - Zero to Hero Series, Infosys Springboard Mar '26

Develop Generative AI Applications: Get Started, Coursera (IBM) Feb '26

Fundamentals of MCP, Hugging Face Feb '26

Natural Language Processing with Classification and Vector Spaces, Coursera Jun '24

VOLUNTEERING

Animal Feeder and Care Taker, Paws Hunger Mumbai
May '22 — Present